ART DUNIYA - Application Knowledge Document

1. App Summary

App Name**: ART DUNIYA**
Version: 1.0
Platform: Web App

Short Description of our App:

Our platform leverages blockchain and smart contracts to deliver a secure, transparent, and values-driven digital rights management system. Creators can protect their work, simplify licensing, and ensure fair royalty distribution with integrity and trust. Every digital asset is immutably recorded on the blockchain, while smart contracts automate ownership validation and payments. With simple API integration, publishers, galleries, and platforms can seamlessly adopt responsible licensing and monetization.

What is ART DRM?

ART DRM (Art Digital Rights Management) is a system that uses technology to protect the ownership, licensing, and usage rights of digital artworks**.** It ensures that artists and creators maintain control over how their work is shared, sold, or licensed online. Unlike traditional DRM, which focuses mainly on preventing copying, ART DRM emphasizes transparency, rightful ownership, and fair compensation often powered by blockchain and smart contracts to provide tamper-proof records and automated royalty distribution.

2. Purpose and Goals

Problems Addressed:

- **Digital Art Theft**

  Over **$600B is lost annually** due to digital piracy.
  75% of artists have experienced their work being stolen or used without permission.

- **Outdated Protection Methods**

  Traditional watermarking and metadata can be easily removed or altered.
  Existing DRM tools are manual, outdated, and not aligned with ethical values.

- **Inefficient Licensing & Lost Revenue**

Artists lose up to 80% of potential revenue due to middlemen and inefficient licensing systems.
Manual approval and tracking processes lead to delays in royalty payments.

- **Lack of Ownership Proof**

  2 out of 3 artists struggle to prove ownership in copyright disputes.
  Secondary sales are not tracked, resulting in an estimated $1.2B in missed resale royalties annually.

Target Audience:

- Faith-conscious & Ethical Digital Artists
- Ethical Content Libraries & Stock Platforms
- Web3 Marketplaces & NFT Platforms
- Publishers & Educational Institutions
- Galleries, Museums & Creative Platforms

Objectives:

- Protect Digital Ownership
- Combat Art Theft & Misuse
- Simplify Licensing Processes
- Ensure Fair & Timely Royalties
- Promote Ethical & Faith-Aligned Practices
- Expand Market Opportunities

3. Key Features and Their Descriptions

Blockchain-Based Ownership Records

Every artwork is immutably recorded on the blockchain, providing undeniable proof of ownership**.**

Automated Smart Contracts

Licensing and royalty payments are executed automatically, ensuring fair, direct, and instant payouts**.**

AI-Powered Piracy Detection

Advanced AI tools monitor digital platforms to detect unauthorized use of artworks

Ethical & Faith-Aligned DRM

A system designed with values-based principles, respecting both legal and moral rights of creators.

Resale Royalty Tracking

Tracks secondary sales to ensure artists continue to receive royalties from future transactions**.**

Seamless API Integration

Easy-to-use APIs allow publishers, galleries, and marketplaces to integrate DRM tools directly into their platforms.

User-Friendly Interface

Simplified dashboards and educational resources to help **artists** adopt blockchain DRM without technical barriers**.**

Multi-Platform Compatibility

Works with digital art, NFTs, stock libraries, educational content, and metaverse ecosystems**.**

4. Ethical Art Marketplace

Our platform includes a **built-in digital art marketplace** where users can:

- **Browse & Discover Art**
  Explore a curated collection of faith-aligned and ethically licensed artworks by global creators.
- **Secure Ownership Proof**
  Each artwork **is** immutably recorded on the blockchain, ensuring authenticity and rightful ownership**.**
- **Fair Licensing & Purchasing**
  Users can license or purchase artworks directly through smart contracts, guaranteeing that artists receive instant and fair payments**.**
- **Search & Filter Options**
  Easily find digital art by category, artist, price, or license type**.**
- **Seamless Transactions**
  Buyers can pay via PayPal (fiat) **or** MetaMask (crypto)**,** ensuring both accessibility and blockchain security.

5. How to Browse Art on the Marketplace

Our marketplace is designed to make discovering and licensing art simple, secure, and enjoyable**.**

Step 1: Access the Marketplace

- Log in to your account and go to the **Explorer** section from the main Navigation Bar.

Step 2: Explore Collections

- Browse featured artworks curated by our team.
- Discover trending art, newly uploaded pieces, or categories like **Photography, Illustrations, NFTs, and Digital Paintings**.

Step 3: Use Search & Filters

- Search by **artist name, artwork title, or keywords**.
- Apply filters for **license type and royalties.**

Step 4: View Artwork Details

- Click on an artwork to see its **preview, details, license terms etc.**

Step 5: Add to Cart / License Directly

- Select whether you want to **purchase, license, or bid** (for NFT artworks).
- Choose your preferred **payment method**: **PayPal (fiat)** or **MetaMask (crypto)**.

Step 6: Complete Transaction

- Once payment is processed, the ownership/license is **instantly recorded on the blockchain**.
- Artists automatically receive **royalties via smart contracts**.

6. Payment Methods

To ensure flexibility, security, and global accessibility, our platform supports the following payment methods:

1. **PayPal (Fiat Payments)**
    - A widely trusted and globally recognized payment gateway.
    - Allows artists, publishers, and platforms to make transactions using traditional currencies (USD, EUR, etc.)**.**
    - Provides buyer and seller protection, making it easy for users who are less familiar with blockchain.
2. **MetaMask (Blockchain-Based Payments)**
    - A secure **Web3 wallet** enabling transactions directly on the blockchain.

- Supports **cryptocurrency payments (ETH, stablecoins, and other tokens)** for licensing and royalty distribution.
- Ensures **transparent, peer-to-peer transfers** without intermediaries, perfectly aligned with our blockchain-powered DRM.

NOTE: There are only two payment methods (i.e., PayPal and MetaMask)

7. Using PayPal for Payments

1. Create a PayPal Account

1. Visit **paypal.com**.
2. Click on **Sign Up**.
3. Select **Personal Account** (for individuals) or **Business Account** (for companies).
4. Enter your **email, name, phone, and password**.
5. Link your **bank account or debit/credit card** for payments.
6. Verify your account via email/SMS.

2. Make a Payment via PayPal

1. On the marketplace checkout page, choose **Pay with PayPal**.
2. Log in with your PayPal credentials.
3. Review the payment amount and recipient.
4. Confirm the transaction.
5. Payment is processed instantly, and **license/ownership is recorded on the blockchain**.

8. Using MetaMask for Payments

1. Create a MetaMask Wallet

1. Install the **MetaMask extension** (Chrome/Firefox) or download the **MetaMask mobile app**.
2. Click **Create New Wallet**.
3. Set a **strong password**.
4. Save your **Secret Recovery Phrase (12 words)** safely this is your backup key.
5. Your wallet is now ready.

2. Add Funds to MetaMask

1. Open MetaMask and copy your **wallet address**.
2. Buy or transfer cryptocurrency (e.g., **ETH or stable coins**) from an exchange like Binance, Coinbase, etc.
3. Funds will appear in your MetaMask wallet.

3. Make a Payment via MetaMask

1. On the marketplace checkout page, select **Pay with MetaMask**.
2. MetaMask pop-up will appear → review gas fee + total amount.
3. Confirm the transaction.
4. The payment is recorded on the blockchain, and **smart contracts instantly distribute royalties** to the artist.

9. License Types

Our marketplace offers flexible licensing options so that creators can protect their work while buyers choose the level of access they need.

1. Viewing License

- Allows the user to view the image online anytime during the license period.
- Supports basic functions such as zoom and rotate.
- Downloading or saving the file is not permitted.
- Suitable for preview, research, or educational access**.**

2. Standard License

- Provides a downloadable protected file on the user's computer.
- Usable in approved applications like Photoshop or other design tools.
- Printing is allowed but may include watermarks or tracking mechanisms.
- Copy/paste or export functions may be restricted or watermarked**.**
- Best for personal use, academic projects, and limited creative work**.**

3. Extended License

- Grants full download access without standard DRM restrictions.
- Allows use in commercial projects (ads, merchandise, publications, etc.).
- Provides higher resolution files for professional applications.
- Comes with fewer usage limitations to support broader creative freedom.
- Ideal **for** publishers, agencies, galleries, and enterprises.

10. App Interface and Navigation

Navigation Structure:

Persistent Features:

Navigation Panel:

Tabs Available (When user is not signed in):

- Home
- About
- Contact
- Faqs
- Explorer
- **SIGN IN**

Tabs Available (When user is signed in):

- Home
- About
- Contact
- Faqs
- Explorer
- Connect Wallet
- Disconnect
- DASHBOARD
- LOGOUT

1. Home Tab

Purpose: The Home tab provides a brief overview of the platform, helping users understand what the system is about.

Includes:

- **Two Buttons:**
  - Get Started
  - Learn More

A. Get Started: On clicking this, the user will be redirected to the Sign In Page.

B. Learn More: On clicking this, the user will be redirected to the Learn More Page.

2. About Tab

Purpose: The About tab provides detailed information about the app, including its vision, mission, objectives, and how it helps creators and users.

3. Contact Tab

Purpose: The Contact tab enables users to reach out by submitting their email address and queries through a simple form, allowing the support team to respond directly within the app.

4. Faqs Tab

Purpose: The FAQs tab provides answers to the most common questions, helping users quickly resolve their queries without contacting support.

Sample FAQS:

If the user asks any question from below or a question similar to this, respond to it according to the answers below.

Q1: What are the features of this platform?
It's a blockchain-powered digital rights management (DRM) solution designed to protect digital art, ensure secure ownership, and provide automated, ethical licensing for creators.

Q2: How does the platform protect my digital art?
Each asset is immutably recorded on the blockchain, making it tamper-proof. Smart contracts handle licensing and ensure your rights and royalties are protected and enforced automatically.

Q3: How are royalties and licensing handled?
Our platform uses smart contracts to automate royalty distribution instantly and fairly — removing middlemen and reducing delays in payments.

Q4: What makes this platform 'ethical'?
The system is built with faith-based values in mind, promoting transparency, fairness, and integrity. It ensures artists retain ownership and are paid fairly without exploitation.

Q5: I'm new to blockchain. Can I still use it?
Absolutely. We provide user-friendly interfaces, onboarding support, and educational materials to help you understand and adopt blockchain without technical expertise.

Q6: How does the platform generate revenue?
Through SaaS subscriptions, transaction fees on licensing, and enterprise-level API integration for marketplaces, educational platforms, and content libraries.

5. Explorer Tab

Purpose: The Explorer tab allows users to browse and discover artworks available on the platform. It provides a visual gallery where all digital assets are displayed with details such as title, artist, category, and license type.

Features of Explorer Tab:

- **Browse All Artworks** → View the complete collection of artworks uploaded by creators.
- **Search** → Find artworks by **title, description, and creator**.
- **Filters** → Find artworks by filtering them based on **License** and **Royalties.**

Features of Individual Cards on Explorer Tab

- **Image**
  - **Location: Center Top**
- **View Details**
  - **Type:** Button
  - **Location:** Bottom left of individual card
  - **Function:** On clicking this button, the user can see all the details of that artwork
- **Buy**
  - **Type:** Button
  - **Location:** Center at the Bottom
  - **Function:** To buy that particular Artwork
- **License**
  - **Type:** Button
  - **Location:** Bottom right of individual card
  - **Function:** To buy the License of a particular Artwork

6. Sign In Tab

Purpose: The Sign In tab allows registered users to securely **log in to their accounts** and access platform features such as uploading art, exploring the marketplace, managing licenses, and tracking payments.

Includes:

- Email field
- Password field
- Login Button
- Continue with Google option
- Forget your password option (In case user forgot the password)
- Signup option
  - Location: Bottom right

7. Connect Wallet Tab

Purpose: The Connect Wallet tab allows users to securely link their **MetaMask** (crypto wallet) account with the platform. This connection is required before making purchases or licensing artworks.

Features:

- **MetaMask Integration** → Connect blockchain wallet to enable **crypto payments** (ETH, stablecoins, etc.) directly on the platform.

- **One-Click Access** → Once connected, users can seamlessly buy, license, or sell artworks.
- **Secure & Encrypted** → Wallet connections are handled with high security, ensuring safe transactions.

8. Disconnect Tab

Purpose: The Disconnect tab provides users with a simple option to safely disconnect their connected wallet **(MetaMask)** from the platform. This ensures privacy and security when the wallet is not in use.

Features:

- **One-Click Action** → Instantly disconnects the linked wallet.
- **Supports MetaMask**→ Works for blockchain payment connections.
- **Enhanced Security** → Prevents unauthorized access to funds or accounts.
- **User Control** → Gives users complete freedom to connect or disconnect anytime.

9. Dashboard Tab

Purpose: The Dashboard tab serves as the **central hub** for users, providing a personalized view of their activities, artworks, licenses, and transactions in one place.

The dashboard contains a Sidebar Navigation Panel which includes:

- Dashboard
- My Artwork
- Upload Artwork
- Licenses
- Piracy Alerts
- Wallet and Royalties
- Settings

9.1 Dashboard

Includes:

- Total Artworks Uploaded
- Total Earnings
- Active Licenses
- Total Licenses Earned
- Active Licensed Detected
- Piracy Cases Detected
- Portfolio Insights

A. Total Artworks Uploaded Panel

- **Location:** Top row, **left column**
- **Function:** Displays Total Artworks uploaded by the user.

B. Total Earnings Panel

- **Location:** Top row, **center column**
- **Function:** Displays the Total Earnings earned by the user by selling Artworks.

C. Active Licenses Panel

- **Location:** Top row, **right column**
- **Function:** Displays all the licenses that users have purchased for your artworks and are currently valid (not expired yet).

D. Total Licenses Earned Panel

- **Location:** Middle row, **right column**
- **Function:** Displays the total number of licenses purchased for your artworks. (active +expired)

E. Active Licenses Detected Panel

- **Location:** Bottom row, **left column**
- **Function:** Displays all the licenses that users have purchased for your artworks and are currently valid (not expired yet).

F. Piracy Cases Detected Panel

- **Location:** Bottom row, **right column**
- **Function:** It lists suspected incidents where an artwork was used without permission, copied, or otherwise misused. It helps creators monitor and respond quickly to potential infringements.

G. Portfolio Insights Panel

- **Location:** Bottom row, **right column**
- **Function:** Displays Artwork Performance and License Analytics, etc.

H. Footer Section

- **Location:** Page Bottom
- **Contents:**
  - <u>ARTDUNIYA</u>
    Empowering Ethical creators with blockchain-powered DRM to protect their digital rights
  - <u>PRODUCT</u>
    1. Artwork Protection
    2. Licensing System
    3. Piracy Detection

4. Pricing

- COMPANY
  1. About us
  2. FAQs
  3. Contact Us
  4. Explorer

9.2 My Artworks

Purpose: Displays all the artworks that a user has uploaded to the platform. It serves as a personal gallery and management area for creators.

9.3 Upload Artworks

Purpose: Allows users to add their Artworks to the platform and make them available for licensing or sale.

Includes:

- Upload
- Details
- Register

A. Upload

- User selects the Artwork to register
- Then click on **CONTINUE TO DETAILS** Button

B. Details

- Enter Artwork Title
- Enter Artwork Description
- Set Price of Artwork
- Set Royalty Percentage
- Then click on **REGISTER YOUR ARTWORK** Button

NOTE: Artwork is only uploaded if it passes all the checks (i.e. AI Detection, Duplicate Detection, Human face Detection)

9.4 Licenses

Includes:

- Manage Licenses
- Issue Licenses

A. Manage Licenses

Includes:

- **Licenses I Hold**
  - Displayed all the Licenses that the user has bought from others for their Artworks

- **Licenses I Granted**
  - Displays all the Licenses that the user has sold to others for his/her Artworks.

B. Issue License

Purpose: Whenever a user uploads his/her Artworks to ARTDUNIYA he/she has to issue a license from ARTDUNIYA

Includes:

A form containing the following fields:

- Select Artwork
- License Wallet Address
- License Type
- Duration(Days)

After that an Issue License Button

9.5 Piracy Alerts

Purpose: The section notifies users whenever the system detects that their artwork is being **misused, copied, or stolen** without permission.

9.6 Wallet and Royalties

Includes:

- Total Royalties
- Artworks
- Transactions
- Available Balance
- Transaction History
- Payment Methods

A. Total Royalties

  - **Location:** Top row, **left column**
  - **Function:** Displays Total royalty generated from the Platform

B. Artworks

- **Location:** Top row, **center column**
- **Function:** Displays the Total number of Artworks uploaded by the user.

C. Transactions

- **Location:** Top row, **right column**
- **Function:** Displays the Total number of Transactions made by user.

D. Available Balance

- **Location:** Middle row
- **Function:** Displays the total balance of your account

E. Transaction History

- **Location:** Bottom row
- **Function:** Displays all the history of transactions made by the user.

F. Payment Methods

- **Location:** Right panel
- **Function:** Displays Connected Wallets and Bank Accounts. It also provides a button of ADD PAYMENT METHOD where the user can add other payment methods (e.g., PayPal)

9.7 Settings

Purpose: Allows users to manage their account security and preferences, ensuring a safe and personalized experience on the platform.

Includes:

- Two-factor Authentication
- Change Password
- Connected Devices

A. Two-factor Authentication

- The user can enable two-factor authentication for secure account login.

B. Change/Update Password

- The user can change/update their password from this tab according to his/her needs

C. Connected Devices

- User can see how many devices their account is logged into and manage them as well.